Employee Attrition

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Introduction

This focus of this project is on a significant issue that all companies face - employee turnover. As a previous software engineer, and now a Software Development Manager, this is a very relevant issue that intrigues me. High employee turnover is a significant financial burden on companies, not to mention the cultural impact. This is more than just the cost of hiring a new employee, since attrition also leads to a loss of institutional knowledge and can affect team morale. In some fields, like software engineering, the loss of a key teammate can have a ripple effect across the team. They may have been the point person on a specific application and if they leave, they take all of that knowledge with them. For higher level employees, like program managers, them leaving may cause a mass exodus of employees who worked for them, or the loss of clients they worked with, especially if they were well respected. Companies need to focus on retaining good employees and understand what causes the most employee upset that drives them to leave a company.

This project will focus on understanding the variables that play into employee attrition and provide managers and HR business partners insight into why employees are leaving companies so they are able to take appropriate actions to retain valuable employees. This project will focus on creating a model that will provide a binary (yes or no) classification for employee attrition. For this project the IBM HR Analytics Employee Attrition & Performance Dataset from Kaggle will be utilized for analysis as well as model training and testing. This is a fictional data set that was created by IBM data scientists to study employee attrition and performance.

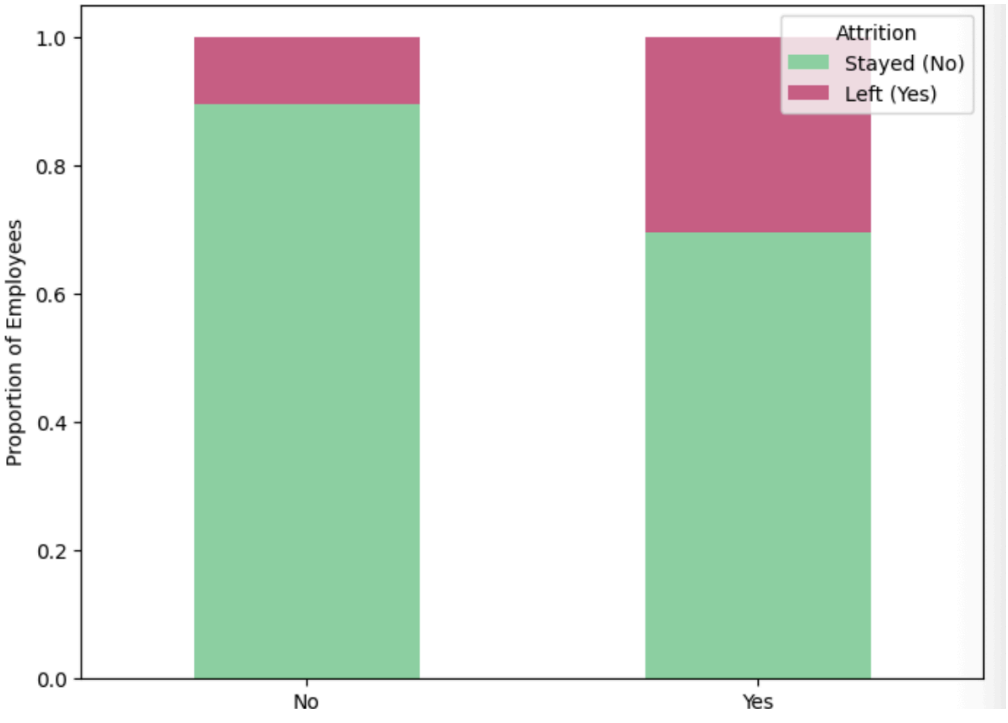
This is a useful and lucrative issue that companies need to solve. Bobbi Kelly, Director-in-Charge for Talent Advisory at Kreischer Miller, states that the cost of replacing an employee ranges from 30% to 250% of their annual salary, depending on role (2024). Beyond the cost for replacing an employee who quits, it can also cause an "emotional and cultural impact" on an entire team, with 52% of employees reporting that a valued coworker quitting negatively affects their own job performance (Staffing Support Specialists, 2025). Managers of teams with high attrition rates also experience emotional stress and become less engaged which affects their ability to lead. Employee attrition comes with multiple levels of costs to companies in separation cost, through pay, benefits, and off-boarding, not to mention the impact to the team. The employee needs to be replaced which adds in recruitment and training costs for a new employee. During the time between an employee leaving and finding a replacement the team will also lose productivity and most likely cause burn out for other teammates due to having to cover work for the employee who left. Solving this issue would allow companies to save time and resources due to having to hire new employees to replace ones who attrit.

This should be pitched to stakeholders and business leaders as a two fold strategy. One, by being able to identify employees who are at risk for attrition and prioritizing stay interviews, bonuses, promotions, and addressing the amount of overtime, the company would be able to save the costs of hiring, training new talent and not lose institutional knowledge. Two, even if the model flags false positives, and additional measures are taken for individuals who look like they are at risk for attrition, but do not attrit, the additional measures and care that the employee would feel from their manager would make them a happier employee. Employees who are happy

and feel appreciated are more engaged and work harder which makes the company more successful.

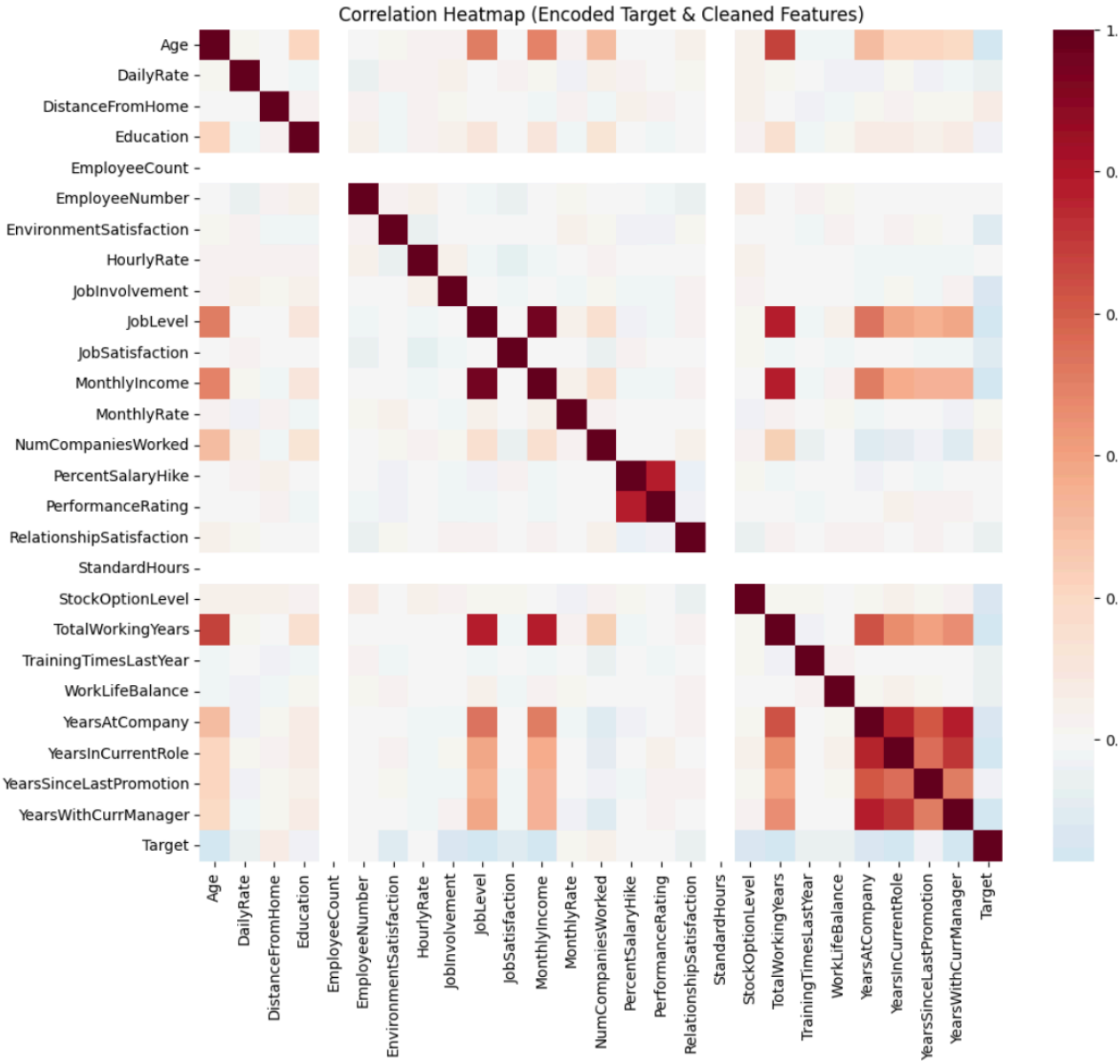
Project Summary

For the Exploratory Data Analysis (EDA) phase the focus was first to gain an understanding of the data. Four graphics were created to help visualize the relationships between the features and the attrition target. The first graph was a Box Plot to show Monthly Income vs Attrition to have a visual queue to gauge if there is a disparity in pay between those who stay at a company and those who leave. This showed that the median monthly income for employees who stayed was visibly higher than those who left. The attrition group was more tightly grouped at lower monthly income levels so the model should be able to perform better for junior to mid-level positions. The second was a Stacked Bar Chart to visualize Overtime vs Attrition to see the effect that overtime has on employees and if it is a factor that contributes to employee attrition. This visualization showed a clear relationship between



Stack Bar Chart: Overtime vs Attrition

overtime and employee turnover. Employees who didn't work overtime had relatively low attrition, however, the proportion of employees who left increased for those who did work overtime. The third was a Correlation Heatmap to view the numeric features to see which variables have the strongest linear relationship with attrition. Before doing the Correlation Heatmap the target variable of attrition was encoded to allow for better assessing the numeric



Correlation Heatmap (Encoded Target & Cleaned Features)

relationships with the target variable. This visualization is especially important to help identify constant or non-informative fields that can be removed to eliminate noise for the model. The heat map identified strong positive correlations between job level, monthly income, and total working years. It also showed that both experience and seniority were primary drivers of compensation, which is to be expected. There was also a negative correlation between *TotalWorkingYears* and the Target that suggests that as tenure increases, the likelihood of attrition decreases. The fourth visualization was a Violin Plot that showed Attrition segmented by Marital Status to visualize if employees are more likely to attrit if they are younger or single compared to people more settled in their careers and are married and older. This visualization showed definite signal regarding age. Single employees had a notably wider band at the younger age groups (20-30s) that suggests that life stage and marital status are significant factors in whether the employee stays or leaves a company. It also showed that single employees had a higher propensity to leave a company compared to married or divorced counterpoints.

Data preparation tasks focused on getting the dataset in a suitable state for the machine learning models. This began with feature pruning and removing fields that would not add beneficial information to the model. The unnecessary columns of *EmployeeCount, Over18, StandardHours, EmployeeNumber,* and *Attrition* were all removed. The first three had constant values across all the rows with no variance, which would provide no data for the model to learn from. The *EmployeeNumber* was a unique identifier for each employee and could lead to overfitting as the model could begin to recognize specific IDs rather than learning the pattern of the data. Finally, since *Attrition* had been encoded that column was dropped to ensure no

duplication in the data so as not to create noise in the model. A check for missing, null, or NaN values was done and noted that there were no missing values in the dataset.

The next step was feature engineering and the creation of a *LoyaltyRatio* to help increase the sensitivity of a model by capturing complex relationships in a single variable. For some employees, especially the older generation, company loyalty is very strong. For other employees they don't have the same company loyalty and move jobs every few years. The value that was being captured here was a sort of tenure density for employees:

$$LoyaltyRatio = \frac{YearsAtCompany}{TotalWorkingYears}$$

The Correlation Heatmap showed *JobLevel*, *MonthlyIncome*, and the *TotalWorkingYears* variables were highly correlated. This will distract the model due to the redundancy of the data, therefore the next step was to consolidate the features. Consolidating these variables into a single variable representing the seniority of an employee retains the vital information that the features provide without causing overfitting since the model will over-emphasize an employee's seniority. The standard Z-Score was captured for each feature and consolidated to represent the new feature: $SeniorityIndex = \frac{z(JL) + z(MI) + z(TWY)}{3}$

This consolidation of features was chosen over something like PCA (Principal Component Analysis) due to PCA being mathematically optimal, but uninterpretable for an HR or manager to understand how the different factors influence an employees longevity at a company. Regularization Lasso (L1) would be able to zero out redundant features, however, it may pick one, like *JobLevel* and discard the others which may have a unique signal for the

model. The *SeniorityIndex* is specifically crafted to allow an HR or business manager to understand the different features of career progression while still maintaining the ability to explain how the different factors accounted for an employee's decision to leave and address the standout factors for why they may leave.

Machine learning algorithms are mathematical models that perform operations like addition and multiplication on numerical vectors. Therefore, the final data preparation task was to perform one hot encoding on the categorical features and create dummy variables to turn these into multiple binary (true/false) values that the model will be able to utilize. This was done with the *drop_first=True* parameter to prevent adding any new multicollinearity into the model. After this, the numeric features were scaled so they would not dominate the model's coefficients

Due to the issue being addressed in this study being a binary one, stay or leave, this dictated the use of a classification model. The focus for model selection was to have two models with different mathematical approaches to validate performance from two different approaches. The two models chosen were Logistic Regression and Random Forest Classifier due to the need to understand which features play an important part in retention and attrition. Models like KNN tend to be less predictive when dealing with human behavior since it is distance-based. Support Vector Machines are powerful, but are black boxes which won't provide the explainability that is needed for this business case. Something like Naive Bayes assumes feature independence and would most likely lead to poor performance since it was already determined that features like *Age*, *JobLevel* and *YearsAtCompany* have a natural relation.

Due to the imbalanced dataset (84/16) the Accuracy score was not going to be the correct choice for validating the model. Instead, the F1-Score was utilized as the primary metric since it

is the harmonic mean of Precision and Recall. It penalizes the model for being "lazy" and just guessing the majority class. Through F1-Score optimization it can be ensured that the model will catch a significant number of at-risk employees without overwhelming the HR department with false alarms. It will also provide a more robust assessment of the model's performance on the minority class than accuracy will. The secondary metric for model validation is Recall (Sensitivity). Recall measures the ability to find the actual at risk employees. It will answer the question: *Out of all the people who actually left, how many did the model correctly catch?* In business, the cost of miss is high. If an integral engineer leaves and the model failed to flag them (False Negative), the company would lose institutional knowledge and face high replacement costs. In HR, the cost of a "False Negative" (failing to identify an employee who is about to quit) is much higher than a "False Positive" (reaching out to someone who was going to stay). Recall measures the model's ability to catch the actual employees who attrit. Recall is a priority as it would be better to have a false alarm and reach out to a happy employee, rather than miss on reaching out to an employee who is truly debating leaving.

Conclusion

Both models naturally performed fairly well at predicting the majority class (employees who stayed) which is natural due to the imbalance of the dataset. The Random Forest model guessed that almost everyone will stay which kept the accuracy value high, while having a low Recall, Precision and F1-Score. The Logistic Regression model outperformed the Random Forest, which missed 43 out of the 47 employees who attrited, and caught an additional 17 at risk

individuals. The first run had lower F-1 Scores which pointed to needing additional tuning and refinement of the model.

For the first refinement threshold tuning was employed to lower the classification threshold from 0.5 to 0.3 which prioritized Recall over Precision. This means if the model is 30% sure an individual will leave the individual would be flagged. In this instance, the model was now catching more than half of the people who actually leave, or 24 out of 47. This is a good improvement over the previous 17 employees who attrited being caught by the model.

The next model update focused on addressing the class imbalance at the algorithm level through setting the *class_weight='balanced'*. This is done during the training phase to focus on addressing beta errors. This version was able to catch 64% of the employees who actually left, or 30 out of 47. The trade off was a lower Precision score which gave 51 false alarms or false positives.

The final model test was using the Synthetic Minority Oversampling Technique (SMOTE) which is an automated, statistical weight to handle imbalanced datasets. This created synthetic data points for the minority class so the model would not focus on, and be biased towards, the majority class. SMOTE was more conservative in Precision at 0.41, which is slightly better than the balanced weights model, and resulted in fewer false alarms. However, this model only caught 49% of the employees who left, or 23 out of 47. Since SMOTE works by drawing lines between existing data points and creating new ones, it is likely that the employees who left are different enough from each other which caused SMOTE to create synthetic data that did not match real behavior.

Model	Technique	Recall	Precision	F1-Score	Accuracy
Baseline	Default Logistic Regression: (0.5)	0.36	0.65	0.47	87%
Tuned	Logistic Regression (0.3)	0.51	0.51	0.51	84%
Refinement 1	Balanced Weights	0.64	0.37	0.47	77%
Refinement 2	SMOTE	0.49	0.41	0.45	81%

Logistic Regression Model Scores

Model	True Negatives (correctly guessed. Would stay)	False Positives (incorrectly guessed they would leave, but they stayed	False Negatives: CRITICAL (incorrectly guessed they would stay but left)	True Positives (correctly identified as 'at risk' of leaving)
Logistic Regression (Default)	238	9	30	17
Random Forest	241	6	43	4
Logistic Regression (Threshold 0.3)	224	23	23	24
Logistic Regression - Balanced Weights	196	51	17	30
Logistic Regression - SMOTE	214	33	24	23

Confusion Matrix Results

The recommendation, moving forward, would be to use the Logistic Regression model with Balanced Weights as a screening tool. It does cast a wide net, but it did a good job capturing most of the employees who did leave, so this would be a place for HR and managers to start. They would have to accept the risk that they would need to filter out false alarms manually.

I would recommend to deploy the model and test it out. I believe in high manager/ employee engagement and believe that having manager check ins and ensuring that employees are happy is better than not pushing for additional measures to gauge employee happiness and to retain valuable talent.

The rate of false negatives, or those 51 employees who were tagged for attrition, but did not attrit is a potential challenge for HR and managers. That is a large number of stay interviews, interventions, or other engagements that HR and managers would take for employees who were not planning on leaving the company. This causes extra work and also costs the company some level of money and needs to be considered. Having additional data, with a less skewed dataset, could be an additional avenue for exploration.

References

HRBench. (2025, August 22). Cost of turnover: How to calculate employee turnover cost in HR. https://www.hrbench.com/resource/learn/cost-of-turnover

Kelly, B. (2024, April 25). The true costs of high employee turnover. Kreischer Miller. https://www.kmco.com/insights/the-true-costs-of-high-employee-turnover/

Staffing Support Specialists. (2025, May 29). The real cost of employee turnover. https://staffing-support.com/real-cost-of-employee-turnover/